Project Negative Lines

Human labeling guide and discrepancy list for the 10 gold-set patents
Prepared 22 October 2026 for the second (human) labeler. Companion files: **Gold set annotations - patents 01-10.xlsx** (labeler A, AI pass), **docs/hand-labeling-guide.md** and **vocab.yaml** v3 in github.com/whatnotalot/negative-lines.

> **If you only read one thing:** nobody expects you to know the "right" answer. Your job is to write down what the patent says, where it says it, and to flag anything that looks odd. A clearly flagged doubt is a perfect label. A silent guess is the only real mistake.

Part 1	How to do the work: setup, order, rules, and how to read each patent
Part 2	Patent-by-patent directions: which pages, tables and paragraphs to open
Part 3	Discrepancy list: 96 items to check one by one, plus 5 cross-patent decisions, with a box to tick and a space for your decision
Part 4	Final self-check before you hand in

Part 1. How to do the work

1.1 Two passes, in this order

1. **Pass 1: label blind.** For each patent, fill in your own records without opening the AI workbook. This is what makes your labels an independent check. If you copy the AI, any AI mistake becomes "truth" and the whole gold set loses its value.
2. **Pass 2: check the discrepancy list (Part 3).** After you finish a patent, go through its items in Part 3. They point at places where the source is ambiguous or contradicts itself. Decide each one and write your decision in the box. It is fine to disagree with what the AI did.
3. **Then compare** with the AI workbook, field by field, and log every difference in the reconciliation log (cause codes in section 1.6).

1.2 Setup (10 minutes)

- Two windows side by side: the patent PDF on the left, your spreadsheet on the right. A second monitor helps a lot.
- Make a copy of the AI workbook, delete all data rows (keep the header row) and save it as `Gold set - labeler B.xlsx`. Use exactly the same sheets and columns so the comparison is easy.
- Keep **vocab.yaml** open for the allowed values of `role`, `metric`, `amount_basis` and `exposure_source`.
- Have this guide printed or on a tablet with a pen, so you can tick boxes.

1.3 Page numbers: two kinds

This guide gives **PDF page** numbers: the page number your PDF viewer shows in its toolbar (page 1 = the first page of the file). Patents also print their own page or column numbers, which are usually different; where useful both are given, for example "PDF p.14 (printed p.13)". Older US patents use **column** numbers (two columns per page) instead of page numbers. US applications number every paragraph like `[0152]`: the fastest way to find one is Ctrl+F for the number.

1.4 The six golden rules (from the hand-labeling guide)

Rule	What it means in practice
1. Copy, don't convert	Type numbers exactly as printed, with the printed unit. "1 min" is 1 with unit min, not 60 s. "0.11 µm" stays 0.11 µm. Code converts later.
2. Missing is empty, never 0	If the patent does not say it, leave the cell blank. 0 means "measured and it was zero".
3. No filling from knowledge	Even if you know SU-8 is baked at 95 °C, only record what this patent states. Put your knowledge in `labeler_note`.
4. Names as written	Copy names and codes exactly, including codenames like "PAG 1", "(A)-1", "DMC". Do not translate them into chemical names unless the patent itself defines them (then put the definition and where you found it in `definition_location`).
5. Every value has a location	Fill `location` (page, table, paragraph or column) and paste a short quote in `evidence_snippet`.
6. When unsure, flag it	Choose your best value, set `labeler_confidence` to low, and write one sentence why in `labeler_note`.

1.5 How to read a patent quickly

Patents are long, but only a small part matters for labeling. In every patent:

1. **Skim the abstract and the first page** to know what kind of resist it is (1 minute).
2. **Skip the long middle section** that lists hundreds of possible compounds and preferred ranges. Ranges like "preferably 1 to 10 parts" are not data. Only come back there to look up what a code (like "MM-1" or "D-2") means.
3. **Go to the section called EXAMPLES** (or "Embodiments"). Read the first worked example in full: it usually gives the recipe and the process.
4. **Then the tables:** one table usually lists the recipes of all examples, another lists the results.
5. **Then the evaluation methods** paragraph, which tells you what each result column means (for example, how "sensitivity" was defined).
6. **Claims at the end:** not data, but sometimes they reveal a typo in the examples (for example a claimed range that the example does not fit).

1.6 Decision rules for the common hard cases

Situation	What to do
Text and table disagree	Record the **table** value, mention the text value in `labeler_note`, set confidence medium, log it with cause `source_conflict`. (Team default; if you disagree, say so, but be consistent.)
Obvious typo ("GPIMS", "[35] 3")	Record exactly what is printed, write the likely intended value in the note, confidence medium.
Amount with no stated basis	Copy number and unit; set `amount_basis` to the "unspecified" value; confidence low.
"wt% relative to resin" / "parts per 100 parts resin"	Basis `parts_per_100_resin`. Plain "parts" of a whole mixture where the resin is not 100: `parts_by_mass`. Grams: `mass_g`.
Same recipe at several doses or thicknesses	One record per dose or thickness (record IDs like `..._ex1_100nm`).
Value only visible in a figure	Do not read numbers off plots. Add a note "figure only".

A table or page is missing or unreadable	Leave the fields empty, confidence low, note "table missing in source copy".
You simply do not understand something	Leave a note and move on. Bring it to the reconciliation meeting. This is normal.

Reconciliation cause codes: `boundary` (records split differently), `basis`, `metric_def` (which metric), `unit`, `missed`, `typo`, `guide_unclear`, `source_conflict`.

1.7 Time budget

Patent	Records (AI)	Difficulty	Suggested time
01 US20210018838A1	6	Easy, 3 pages. Start here.	30 min
04 US20020068237A1	6	Medium (amount bases)	45 min
05 KR20240036456A	8	Medium (cut-off print, Korean table)	45 min
10 US20070248896A1	46	Medium (many small result tables)	60 min
08 US9017923B2	16	Medium	45 min
09 US20230314945A1	19	Medium (codes)	60 min
07 US7432034B2	17	Hard (codes, 67 pages)	75 min
06 US20080241753A1	49	Medium (repetitive)	60 min
02 WO2004107052A1	51	Easy but repetitive	45 min
03 CN120370628A	21	Hard: tables missing	30 min, then stop

Tip: copy-paste rows for repetitive examples (02, 06, 10) and change only what differs. Take a 5-minute break between patents.

Part 2. Patent-by-patent directions

For each patent: where the recipe, the process and the results are, and what to watch out for. "PDF p." = page number in your PDF viewer.

01. US20210018838A1, negative photoresist for semiconductor encapsulation (file "01.US20210018838A1.pdf")

3 PDF pages. 6 examples. Coating thickness and uniformity only (no exposure, no development).

What	Where
Ingredient definitions	PDF p.2 (printed p.1): [0015], [0018]-[0021]; [0022] solvent oils
Recipes	PDF p.3: Table 1 (left column). Amounts in wt%, leveling agent in ppm.
Process	PDF p.3: [0023] and [0025] (they disagree, see Part 3)
Results	PDF p.3: Table 2 (left bottom) and Table 2-continued (right top): thickness at Center, 1/2, 1/4, 1/8, Edges, Average, and u %

02. WO2004107052A1, negative photoresist for spinless (slit) coating

24 PDF pages. 51 embodiments. Coating quality only (no exposure, no development).

What	Where
Base recipe for all embodiments	PDF p.12 (printed p.11), lines 9-21: "Embodiments 1 ~ 19"
Embodiments 20-51 change	PDF p.13 (printed p.12), lines 1-4: DMC as the only solvent; surfactant per Table 2
Solvent list and abbreviations	PDF p.11 (printed p.10), lines 9-23
Surfactant trade names	PDF p.11 (printed p.10), lines 1-5
Test methods	PDF p.13-14 (printed p.12-13): coating property, uniformity, edge profile
Table 1 (Emb. 1-19)	PDF p.14 (printed p.13)
Table 2 (Emb. 20-51)	PDF p.15-16 (printed p.14-15)

03. CN120370628A, negative photoresist composition (Google Patents English print)

7 PDF pages. Tables 1 and 2 are missing from this copy.

What	Where
Resin syntheses (A1-A4, B1-B4)	Examples 1-8, around PDF p.4-5
Recipe W1	Example 9, PDF p.5
Recipes W2-W20	Table 1: **missing** (caption only)
Process	Example 21 (PDF p.5), test methods Example 33 (PDF p.5-6), developing time in claim 9
Results	Table 2: **missing**. Only group statements in the text (PDF p.6)

> **Recommendation:** label only Example 21 (recipe W1) and the comparative examples' text statements, then stop. Ask the team to find the original Chinese publication with the tables before spending more time.

04. US20020068237A1, negative photosensitive resin composition and dry film

6 examples, visible-light exposure on different substrates. Use Ctrl+F for paragraph numbers.

What	Where
Basis of all amounts	[0151]: "parts and percentages are all by weight"
Example 1	[0152]-[0154]
Example 2	[0155]-[0157]
Example 3	[0158]-[0160]
Example 4	[0161]-[0163]
Example 5	[0164]-[0166]
Example 6	[0167]-[0169]

Each example has three parts: the recipe (first paragraph), a flood-exposed test of the cured coating (pencil hardness, solvent or chemical resistance), and a patterned test through a mask. Note which result belongs to which test.

05. KR20240036456A, negative photosensitive resin composition and dry film resist (Google Patents English print)

14 PDF pages. 5 examples + 3 comparatives. The right edge of most lines is cut off in this print.

What	Where
Novolak resin syntheses A-1 to A-6	PDF p.8-9
Example 1 recipe; Examples 2-5, Comparatives 1-3	PDF p.9
Four evaluation methods	PDF p.9 (bottom) to p.10
Table 1 (results)	PDF p.11, printed sideways with Korean headings (turn your head right, or rotate the page)
Full uncut text	patents.google.com/patent/KR20240036456A/en (open in a browser)

Korean headings in Table 1: 실시예 = Example; 비교예 = Comparative example; 성분(A) = component (A); 구조단위의 몰비 (a1)/(a2)/(a3) = molar ratio of structural units; 반응 용제 = reaction solvent (에탄올 ethanol, 프로판올 propanol); 제막성 = film forming ability; 알칼리 용해성 = alkali solubility; 밀착성 = adhesion; 경화막의 내약품성 = chemical resistance of cured film. ○ = good, × = insufficient.

06. US20080241753A1, negative resist composition (silsesquioxane, e-beam)

16 PDF pages (PDF page = printed page + 3). 24 examples, each at two resist thicknesses, + 1 comparative.

What	Where
Resin synthesis (X1)	PDF p.12 (printed p.9): [0147]-[0150]
Example 1 recipe and two-layer process	PDF p.12-13: [0151]-[0162]

Example 2 single-layer process	PDF p.13: [0163]-[0167]
Which examples use which method; code meanings	PDF p.13: [0168]-[0171]
Table 1 (recipes)	PDF p.13-14 (printed p.10-11)
Table 2 (Examples 1-13 results)	PDF p.14 (printed p.11)
Table 3 (Examples 14-24, Comparative 1)	PDF p.15 (printed p.12)

07. US7432034B2, negative resist composition (Fujifilm, e-beam and X-ray)

67 PDF pages, but you only need about 8. Older patent: two numbered columns per page; PDF page = (column + 3) / 2.

What	Where
What the invention claims	PDF p.2 (columns 1-2)
Code lists (resins (1)-(95), MM-x, CL-x, PAG-x, J-x)	Middle pages, roughly PDF p.4-34. Only look up codes you need.
Synthesis of Resin (29), MM-1, acid generator I-1	PDF p.35-37 (columns 67-71)
Example 1 recipe, process and test methods	PDF p.37 (columns 71-72)
Table 1 (recipes) and quencher codes D-1 to D-3	PDF p.38 (column 73)
Table 2 (e-beam results) and Table 3 (X-ray results)	PDF p.39 (column 75)

08. US9017923B2, negative resist with ferrocene polymers (Shin-Etsu, e-beam)

PDF page = (column + 3) / 2.

What	Where
General process ranges (not data)	PDF p.19 (column 35-36)
Polymer syntheses 1-6 and comparative polymers	PDF p.19-22 (columns 36-41)
PAG 1, Amine 1, CR 1 structures; solvents; e-beam process	PDF p.22 (column 42)
Table 1 (recipes and results)	PDF p.23 (column 43)
Dry etching test and Table 2	PDF p.23 (columns 43-44)

09. US20230314945A1, negative-tone resist with silicon resin (Tokyo Ohka, EUV)

PDF page = printed page + 1.

What	Where
Table 1 (Examples 1-9) and Table 2 (Examples 10-17, Comparatives)	PDF p.38-39 (printed p.37-38)
Meaning of codes; resin Mw and ratios	[0481]-[0488], PDF p.39
Structures of A-x, B-x, C-x, D-x	PDF p.39-42
(E)-1, (S)-1, (S)-2; process steps	[0504]-[0523], PDF p.42-43
Table 3, Table 4 (Eop, LWR) and Table 5 (etch)	PDF p.43 (printed p.42)

10. US20070248896A1, SU-8 photoresist for high aspect ratio plating (FormFactor)

6 PDF pages (PDF page = printed page + 1).

What	Where
Ingredient roles and general process	PDF p.3-4: [0017]-[0035]
Examples 1 and 2 (grams)	PDF p.4: [0038]-[0044]
Examples 3-6 (wt%)	PDF p.5: [0045]-[0048]
Example 7 adhesion, Tables I and II	PDF p.5: [0049]-[0052]
Example 8 via cracking (Table III) and wafer bow (Table IV)	PDF p.6: [0053]-[0057]

Part 3. Discrepancy list

Work through these after your blind pass of each patent. For each item: open the place given, read it yourself, tick the box, and write your decision (for example "agree with AI", "use text value", "new record", "ask team"). "AI did" shows what labeler A recorded, so you can see where you differ.

01. US20210018838A1

Done	ID	Where to look	Issue	AI did	Your decision
☐	01-1	File name	File was named "01.Invoice-8GG0IQDD-0002.pdf" but contains this patent. Renamed to 01.US20210018838A1.pdf on 22 Oct 2026; nothing to decide.	Labeled as US20210018838A1	
☐	01-2	PDF p.3, [0023] vs [0025]	[0023]: wafer baked at 120 °C for 10 min, coated 110 µm. [0025]: coating baked at 110 °C for 5 min after spin coating. Two different bakes.	[0025] as softbake; [0023] bake as pre-treatment	
☐	01-3	[0023] vs Table 2, [0027]	Text says 110 µm coating; Table 2 measures about 49-66 µm; [0027] says "about 50 um".	film_thickness empty; measured values as results	
☐	01-4	[0027] vs Table 2	Text claims uniformity below 5%, but Example 1 is 9.0% and Example 6 is 5.0%.	Recorded table values	
☐	01-5	Table 1	Resin + photosensitizer + solvent already sum to 100 wt%; leveling agent is in ppm on top, basis not stated.	ppm, basis unspecified	
☐	01-6	[0021]	"Photosensitizer" is a sulfonium hexafluoroantimonate + diazonaphthoquinone mixture (chemically a photoacid generator).	Role sensitizer, as the source says	
☐	01-7	[0020]	Resin Mw only as a class range (80,000-140,000), not per example.	Range in note, Mw empty	
☐	01-8	Table 2 header	"u %" formula is not defined.	Recorded as printed	

02. WO2004107052A1

Done	ID	Where to look	Issue	AI did	Your decision

☐	02-1	PDF p.14, Table 1; PDF p.11 lines 9-23	Solvent codes DMC, NBA, MCA, CHX, PGPE are never defined. Only PGMEA and EEP are spelled out.	Kept codes, low confidence	
☐	02-2	Table 1 "ratio% (I:II)"	Not stated whether the solvent ratio is by mass or volume.	Basis unspecified	
☐	02-3	PDF p.15-16, Table 2 "content"	Surfactant content 0.01 or 2 has no unit. Base recipe uses 0.03 wt% (PDF p.12 line 14); description uses "parts by weight" (PDF p.11 line 1).	Unit empty, basis unspecified	
☐	02-4	PDF p.13 lines 1-4	Emb. 20-51: DMC is the only solvent, and the surfactant replaces R08. Easy to miss.	DMC single solvent, Table 2 surfactant	
☐	02-5	PDF p.13-14 test methods	No exposure or development: results are coating quality only. Decide whether these records belong in the lithography dataset (flag `coating_only`).	Noted, no flag (v2)	
☐	02-6	PDF p.12 line 17-21	Resin is "general formula 1 (X methyl, Y1 methyl, Y2 formula (II) with R1 methyl)": the formula is earlier in the document; no Mw given.	Name as written, no Mw	
☐	02-7	Table 2 row 21; PDF p.11 line 3	Minor: "Good" vs "good"; "F177P" in text vs "F177-P" in table.	Copied as printed	
☐	02-8	PDF p.12 line 21 and p.13 line 3	Viscosity "about 4 cps" is a solution property, not a lithography result.	Result metric other	

03. CN120370628A

Done	ID	Where to look	Issue	AI did	Your decision
☐	03-1	Captions "Table 1", "Table 2"	Both tables are missing (only captions). Recipes W2-W20 and per-example results are unknown.	Empty fields, low confidence	
☐	03-2	Example 9 vs claims	Recipe W1 uses 1 g of resin B, but the claims ask for 10-20 parts. Possible translation error.	Recorded 1 g, low confidence	
☐	03-3	Example 9	Not stated which of A1-A4 and B1-B4 is used.	"resin A", "resin B"	

☐	03-4	Example 33 (3)	Dose "30-70 mJ": a range, no /cm², in the general method, not per example. Example 21 gives exposure time 15 s.	Dose as text range, time 15 s	
☐	03-5	Note below Table 2	Grade symbols for adhesion and edge roughness are garbled in translation.	Scale text without symbols	
☐	03-6	Example 21	"Baked and photo-cured at 120 °C for 60 s" after exposure: post-exposure bake or cure?	Recorded as PEB	
☐	03-7	Claim 9	Developing time 60 s appears only in a claim, not in the examples.	Used claim value	
☐	03-8	Comparative example 9	Only the main resin (polyisoprene) is named; rest of recipe unknown.	One component	
☐	03-9	Text after Example 33	Results for Examples 21-32 are one group statement ("up to 1.5 µm resolution, good adhesion"). Should a group statement be copied to each record?	Copied to each record	

04. US20020068237A1

Done	ID	Where to look	Issue	AI did	Your decision
☐	04-1	[0151]	Amounts are "parts by weight" of the whole mixture; the resin is 100-167 parts, so this is not "per 100 parts resin".	Basis parts_by_mass	
☐	04-2	[0152], [0155], [0158], [0161], [0164], [0167]	Resins are given as solutions or varnishes (nonvolatile 50-70%). The amount is the solution, not the dry resin.	Solution amount + note	
☐	04-3	[0152]	Example 1 resin solution solvent "xylol/butanol (50/50)": separate solvent components or part of the resin?	Two solvent components	
☐	04-4	[0158], [0161], [0164], [0167]	Melamine resins act as crosslinkers chemically, but the patent calls them resins.	Role resin + note	
☐	04-5	All examples	"Photoacid proliferating agent": new role `acid_amplifier` in vocab v3.	Role other (v2)	

☐	04-6	All examples	Exposure: filtered metal halide lamp, visible light ≥480 nm. Use `visible` + wavelength_min_nm 480 (v3).	Source other (v2)	
☐	04-7	[0163]	Example 4's patterning step does not mention a mask ("same amount of the same visible light").	Noted, medium confidence	
☐	04-8	Each example	Pencil hardness and resistance are measured on a flood-exposed coating (no mask), a different test from the pattern. Same record or separate?	Same record, condition noted	
☐	04-9	Each example	Softbake only "dried", no conditions. Heat after exposure is "for curing".	Softbake empty; cure as PEB	
☐	04-10	Each example	Photosensitizer is only drawn as a structure, no name.	Described as structure	

05. KR20240036456A

Done	ID	Where to look	Issue	AI did	Your decision
☐	05-1	Whole print	Right edge of most lines is cut off. Read the missing words on the Google Patents web page.	Filled from web page	
☐	05-2	PDF p.9 Example 1	Check: 10.0 g resin, 6.0 g DPHA (Nippon Kayaku), 1.2 g Irgacure369, 25.8 g PGMEA (the 25.8 g is cut off in the print).	As listed	
☐	05-3	PDF p.9-10 evaluations	Four tests on three substrates (PET, copper, silicon) with different processes. One record or four?	One record per formulation	
☐	05-4	Chemical resistance test	200 mJ/cm² exposure and 130 °C 180 s bake apply only to this test; TMAH dip 10 s is for the unexposed alkali test.	All in one process row + note	
☐	05-5	PDF p.8-9 syntheses	Mw of A-2 to A-4 cut off (web page: 2,370 / 2,700 / 2,900); A-6 Mw not legible.	Web values; A-6 empty	
☐	05-6	Comparative Example 2	Resin A-7 is only a formula image.	Code only, low confidence	
☐	05-7	Comparative Example 3	No component (A) at all.	No resin component	

☐	05-8	Table 1 row 2	"1/0.61/0.4" etc. are molar ratios of monomers inside the resin, not formulation amounts.	In polymer_monomer_ratio	
☐	05-9	Film forming test	Thickness "about 5 µm"; bar coater "No. 02".	5 µm, as_coated	

06. US20080241753A1

Done	ID	Where to look	Issue	AI did	Your decision
☐	06-1	[0158], [0163] vs Table 2 (PDF p.14)	Text: Example 1 = 50/100 nm, Example 2 = 100/200 nm. Table 2: the reverse.	Used text values (against team default!)	
☐	06-2	[0162], [0167]	Solid content per thickness given only for Examples 1 and 2.	Filled for 1, 2 only	
☐	06-3	Tables 2-3	Each example at two thicknesses with identical results. Two records per example?	Two records each	
☐	06-4	[0152] and Table 1	"Resin 1" is X1 only by inference from Example 1. "Resin 2" (Comparative 1) is never defined.	Resin 1 = X1 (medium); Resin 2 low	
☐	06-5	Table 1 rows 19-24	"PAG3:PAG4 = 95:5 (10 parts)": a mixture with a total and a ratio.	Two PAG rows, text amount	
☐	06-6	[0158], [0164]	No e-beam dose given anywhere.	Dose empty	
☐	06-7	[0161], [0171]	"Resolution in L/S 50 nm": line width or half-pitch?	Value 50, medium confidence	
☐	06-8	[0168]	Odd examples use the two-layer method (Example 1), even ones single-layer (Example 2). Easy to swap.	Followed [0168]	
☐	06-9	Table 1 rows 13-14	Crosslinker 12 parts (others 10 or 7.3). Real or typo?	Recorded 12	
☐	06-10	[0159]	"Dried with heating at 100 °C for 60 s" after rinse: hard bake?	Recorded as hardbake	
☐	06-11	[0170]	PAG1 (triflate) and PAG2 (nonaflate) are fluorinated sulfonates: relevant for the compliance agent later.	Not flagged	

07. US7432034B2

Done	ID	Where to look	Issue	AI did	Your decision
☐	07-1	PDF p.36 (col. 69) vs Table 1 (PDF p.38)	Resin (29): synthesis says Mw/Mn 2.2, Table 1 says 1.2.	Table value 1.2	
☐	07-2	Table 1: Example 7 vs Comparative 2	Resin (31) listed with Mw 8000 and with Mw 9200.	Copied each as printed	
☐	07-3	Table 1, Example 10	Crosslinker "C-3" while all others are "CL-n": probably CL-3.	As printed, low	
☐	07-4	Example 1 (PDF p.37)	Solvent (7.5 g PGMEA + 1.5 g PGME) and surfactant W-1 (0.001 g) are stated for Example 1 only. Assume the same for 2-12?	Assumed same	
☐	07-5	Col. 72 "(2-4) Line Edge Roughness"	Called LER but measured as 3σ of the line width (which is LWR).	ler_3sigma_nm + note	
☐	07-6	Table 2 vs Table 3	Sensitivity in μC/cm² for e-beam but mJ/cm² for X-ray. Never convert.	Units kept	
☐	07-7	Tables 2-3	Resolution printed in μm (0.11), not nm.	Kept μm	
☐	07-8	Col. 75 "Patterning by X-Ray"	Example 13 and Comparatives 3-4 reuse the recipes of Example 1 and Comparatives 1-2.	Recipes copied	
☐	07-9	Table 1	Codes (resin numbers, MM-x, CL-x, I-1, PAG4-x, J-x) are defined in the long code lists; AI did not look up resin structures.	Codes only	
☐	07-10	Col. 72 "(2-1) Sensitivity"	Sensitivity = dose to resolve 0.15 μm L/S 1:1, so metric `sensitivity_esize`.	sensitivity_esize	
☐	07-11	Col. 71; PAG4-36 structure	I-1 (nonaflate) and PAG4-36 (C4F9SO3) are PFAS-type acid generators: compliance relevance.	Not flagged	

08. US9017923B2

Done	ID	Where to look	Issue	AI did	Your decision
☐	08-1	PDF p.22 col. 42 vs Table 1 (PDF p.23)	Text: PEB 110 °C for 90 s. Table 1 "PEB temp.": 95 °C.	Table value 95 °C	
☐	08-2	Col. 42	"6 inch (200 mm)" substrate: 6 inch is 150 mm.	Copied, noted	

☐	08-3	Synthesis Example 6; Table 1 Example 1-6	Polymer 6 has the acid generator built into the polymer, so no separate PAG.	No PAG component	
☐	08-4	Table 1	Examples 1-3 and 1-5 have no crosslinker ("—").	No crosslinker	
☐	08-5	Col. 42 last paragraph	Text: "edge roughness of a 120-nm LS"; table header: LWR. Sigma not stated.	lwr_3sigma_nm, medium	
☐	08-6	Col. 42 headings	Section titled "Examples 1 and 2", table rows "Example 1-1 … 1-6", text says "Examples 1 to 6".	Used table labels	
☐	08-7	Table 2 (etching)	Etch rates are for polymer-only films, not resist recipes. Separate records?	8 separate etch records	
☐	08-8	Table 1 solvents	Solvent amounts like 2000 parts per 100 polymer: copy as printed.	parts_per_100_resin	

09. US20230314945A1

Done	ID	Where to look	Issue	AI did	Your decision
☐	09-1	Table 2-continued, Example 15 (PDF p.39)	Quencher printed "(D)-3 [35] 3".	Recorded 35, medium	
☐	09-2	Same table, Example 13	S-2 printed "4800" without brackets.	Recorded 4800	
☐	09-3	[0481]	Amounts are "parts by mass: in terms of solid content". Solvent amounts are also in parts.	parts_per_100_resin	
☐	09-4	[0480]	Solid content "about 0.70% by mass" (approximate).	0.70	
☐	09-5	[0519]; Table 2 Example 17	Example 17: organic developer (butyl acetate), no rinse, different crosslinker amount (30) and solvent amounts.	Recorded differences	
☐	09-6	[0520]-[0525]	Only one size was tested (16 nm line, 32 nm pitch). Is that a "resolution"?	Resolution 16 nm, "smallest tested"	
☐	09-7	[0522], [0531]; Table 4	Comparative 1: not resolved; film fully dissolved. No Eop or LWR.	Qualitative result only	
☐	09-8	[0533]-[0538], Table 5	Etch test uses a different film (50 nm, 90 °C 60 s) and only Examples 1, 3, 12.	Added as extra result with condition	

☐	09-9	[0499]-[0503]	(D)-x are photodecomposable bases (sulfonium salicylates), role quencher.	quencher	
☐	09-10	B-1 to B-7 structures	All PAG anions are fluorinated sulfonates: compliance relevance.	Not flagged	

10. US20070248896A1

Done	ID	Where to look	Issue	AI did	Your decision
☐	10-1	[0045] Example 3	"GPIMS" printed; everywhere else GPTMS.	As printed, medium	
☐	10-2	Table III header vs [0035]	Dose printed "MJ/cm2" (megajoule); [0035] says "500 to 2500 mJ". Almost certainly mJ/cm^2.	mJ/cm^2 + note	
☐	10-3	Table II (PDF p.5)	**Suspicious duplicates:** die (1,1) = (1,6) = 59.02 (2.15); (8,1) = (8,6) = 64.64 (3.61); (16,1) = (16,6) = 66.99 (4.85). Likely a copy error in the patent.	Copied as printed, not flagged	
☐	10-4	Table II	Typos: "66.99 (4.85" missing ")"; "Waver average".	Ignored	
☐	10-5	[0050] vs Tables I-II	Text rounds: 49.5 / 63.6 / 51.3 / 61.2 vs tables 49.5 / 63.55 / 51.32 / 61.21.	Table values	
☐	10-6	[0049], [0053]	"Unmodified SU-8" recipe is not restated (probably Example 1).	Component without amount, low	
☐	10-7	Example 2 vs Example 3	Example 2 in grams works out to 62.4 / 26.2 / 1.0 / 6.4 / 4.0 %; Example 3 says 62.4 / 26.3 / 6.3 / 1 / 4. Rounding or a different recipe?	Separate records	
☐	10-8	[0045]-[0048]	Examples 3-6: "coated, dried, exposed, PEB" but no conditions and only qualitative claims.	Records with qualitative claim	
☐	10-9	Table III	Two rows per dose and cooling: assumed two wafers. One record per row?	One record per row and formulation	
☐	10-10	Table IV	Wafer bow has no unit; two columns both called "Dot"; "Before coat" Dot 2 is NA.	Named Dot 1 / Dot 2, unit empty	
☐	10-11	[0022]	Cyracure 6974 is called "photoinitiator" but is a triaryl sulfonium SbF6 salt.	Role pag + note	

☐	10-12	[0027]-[0028]	DOP is a plasticizer; vocab has no plasticizer role.	Role additive + note	

Cross-patent items (decide once, apply everywhere)

- ☐ **X-1** Text vs table conflicts (01-2, 06-1, 07-1, 08-1): the AI was not consistent (06-1 used text, others used tables). Agree on one rule.
- ☐ **X-2** Coating-only patents (01, 02, parts of 05, 08 etch, 10): keep in the gold set but flag `coating_only`?
- ☐ **X-3** Group statements copied to many records (03-9, 10-8): allowed or not?
- ☐ **X-4** Missing vocab values: X-ray source (07), plasticizer role (10), polymer-bound PAG (08). Note them for vocab v4.
- ☐ **X-5** PFAS-type acid generators (06, 07, 08, 09): not part of labeling, but tell the compliance owner.

Part 4. Final self-check before you hand in

Done	Check
☐	Every record ID follows `<doc_id>_<example label>` and comparatives are marked `is_comparative = TRUE`.
☐	Every record has `location` and a short `evidence_snippet`.
☐	Every component has a `role` and `name_as_written` (codes kept as printed).
☐	Every amount has `amount_raw`, its printed unit, and an `amount_basis`.
☐	Doses keep their printed unit (mJ/cm^2, µC/cm^2); nothing was converted by hand.
☐	Each sensitivity value has the right metric (Esize / Eop, gel dose, D50) and its wording in `metric_as_written`.
☐	Missing values are empty, not 0. No values from memory or from plots.
☐	Everything you were unsure about has confidence low or medium and a one-line note.
☐	All Part 3 items have a tick and a decision.
☐	You only then opened the AI workbook and logged each difference with a cause code.

> **Remember:** disagreements with the AI are useful findings, not mistakes. The point of a second labeler is to find exactly those places. If two careful people disagree, the guide needs a clearer rule, and that is a result too.